കണ്ണൂർ ജില്ലാ പഞ്ചായത്ത്

പൊതുവിദ്യാഭ്യാസ
വകുപ്പ് - കേരളം

ഡയറ്റ് കണ്ണൂർ

समग्र शिक्षा के रल
Samagra Shiksha Kerala
സമഗ്ര ശിക്ഷാ കേരളം



എസ് എസ് എൽ സി
വിദ്യാർത്ഥികൾക്കുള്ള പഠനപിന്തുണാ സഹായി

2

കെമിസ്ട്രി

ENGLISH MEDIUM

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കണ്ണൂർ ജില്ലാപഞ്ചായത്ത് - പൊതുവിദ്യാഭ്യാസവകുപ്പ്
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തയ്യാറാക്കിയത്

സ്മൈൽ റിസോഴ്സ് ഗ്രൂപ്പ് അംഗങ്ങൾ

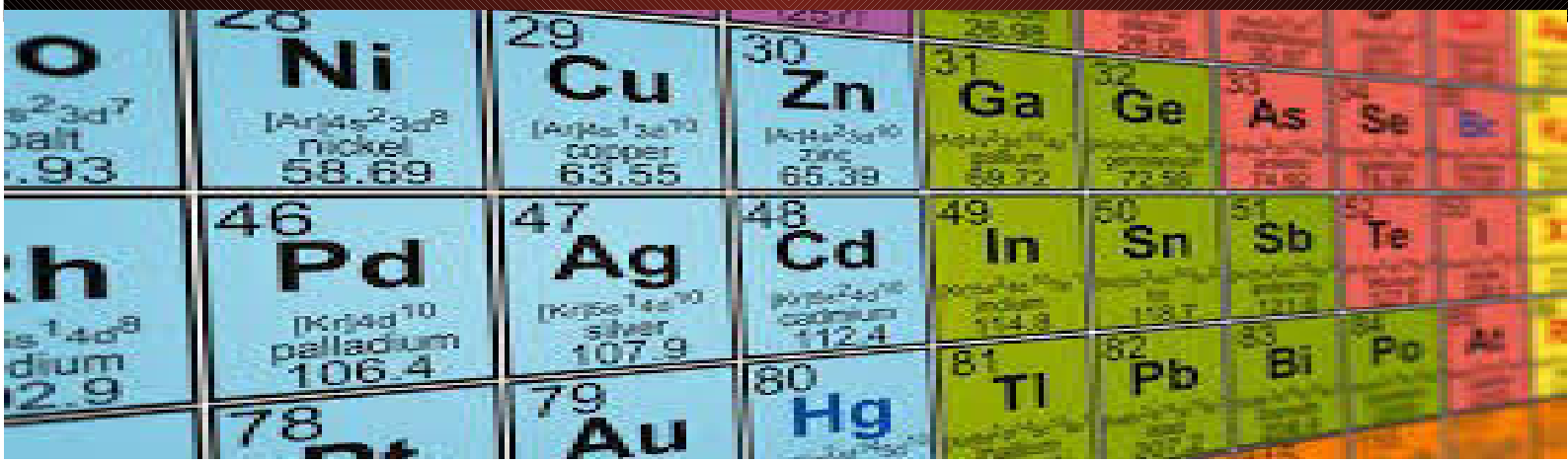
ഏകോപനം

ജില്ലാവിദ്യാഭ്യാസപരിശീലനകേന്ദ്രം ഡയറ്റ്, കണ്ണൂർ

SMILE

എസ് എസ് എൽ സി പഠന സഹായി

2025-26



CHEMISTRY



ജില്ലാ പഞ്ചായത്ത് - കണ്ണൂർ

ഡയറ് കണ്ണർ

ജില്ലയിലെ വിദ്യാഭ്യാസ മേഖലയിൽ ഗുണപരമായ മികവ് വളർത്തിയെടുക്കുന്നതിന് കണ്ണൂർ ജില്ലാപഞ്ചായത്ത് നടപ്പിലാക്കിക്കൊണ്ടിരിക്കുന്ന സമഗ്ര വിദ്യാഭ്യാസ പദ്ധതിയാണ് " SMILE ". എസ് എസ് എൽ സി പരീക്ഷയിൽ നമ്മുടെ ജില്ലയെ ഒന്നാം സ്ഥാനത്തേക്ക് ഉയർത്തുന്നതിൽ ഈ പദ്ധതി ഒരു പ്രധാന പങ്ക് വഹിച്ചിട്ടുണ്ട്.

കേവലം ചോദ്യങ്ങൾ എന്നതിലുപരി പഠന പ്രവർത്തനങ്ങൾക്ക് സാധ്യത നൽകാൻ രസതന്ത്രത്തിൽ ശ്രമിച്ചിട്ടുണ്ട്. ഭിന്ന നിലവാരത്തിലുള്ള കുട്ടികളെ പരിഗണിച്ചാണ് രസതന്ത്രം " SMILE " തയ്യാറാക്കിയിട്ടുള്ളത്. പുതിയ പാഠപുസ്തകത്തിന്റേയും ചോദ്യപേപ്പർ ഘടന മാറിയതിന്റേയും പ്രയാസങ്ങൾ ലഘൂകരിക്കാനുതകുന്ന പ്രവർത്തനങ്ങൾ നൽകിയിട്ടുണ്ട്. കണ്ണൂർ ഡയറ്റിന്റെ നിർദ്ദേശങ്ങളും അക്കാദമിക ഇടപെടലും ഏറെ ഉപയോഗപ്പെടുത്തിയിട്ടുണ്ട്.

രസതന്ത്രം ടെക്സ്റ്റ് ബുക്കിനൊപ്പം അദ്ധ്യാപകർക്കും കുട്ടികൾക്കും ഉപയോഗിക്കാൻ കഴിയുന്നതാണ് രസതന്ത്രം " SMILE" . കൂട്ടായ്മയിലൂടെ മുഴുവൻ കുട്ടികളെയും മികച്ച പഠന നിലവാരത്തിലേക്ക് എത്തിക്കേണ്ട ഉത്തരവാദിത്തം ഏറ്റെടുത്തുകൊണ്ട് ജില്ലാ പഞ്ചായത്തിന്റെ ഈ സദുദ്യമത്തെ പൂർണ്ണതയിലെത്തിക്കാൻ നമ്മുക്ക് സാധിക്കട്ടെ.

രസതന്ത്രം " SMILE" ശില്പശാലയിൽ പങ്കെടുത്തവർ

മനോജ് പി എം
എച്ച് എസ് ടി
മമ്പറം എച്ച് എസ് എസ്

കെ ഉണ്ണികൃഷ്ണൻ
എച്ച് എസ് ടി
കടമ്പൂർ എച്ച് എസ് എസ്

ഡോ: സൂരജ് എം പി
എച്ച് എസ് ടി
എസ് എസ് ജി എച്ച് എസ് എസ്
പയ്യന്നൂർ

സംഗീത ബാലകൃഷ്ണൻ സി ആർ
എച്ച് എസ് ടി
ചെമ്പിലോട് എച്ച് എസ് എസ്
തലവിൽ

CHEMISTRY

PART - 2

Standard

X

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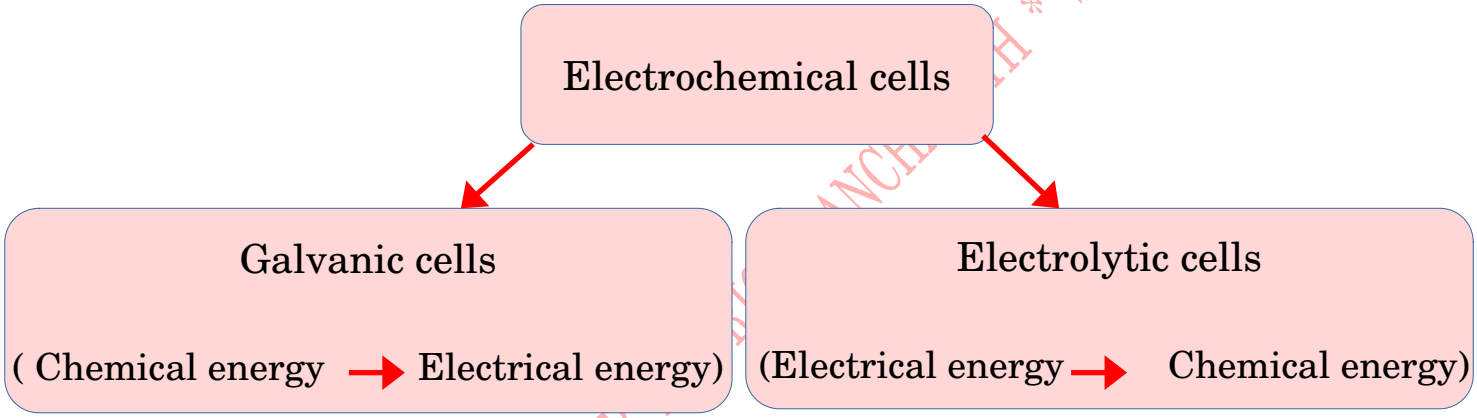
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5 ELECTROCHEMISTRY

Electrochemistry is the branch of chemistry which deals with the studies on the processes that produce electricity through chemical reactions and use electricity to bring about chemical reactions.

Electrochemical cells are the devices that make such changes possible. Electrochemical cells can be divided into two types.

1. Galvanic cells
2. Electrolytic cells



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Galvanic cells

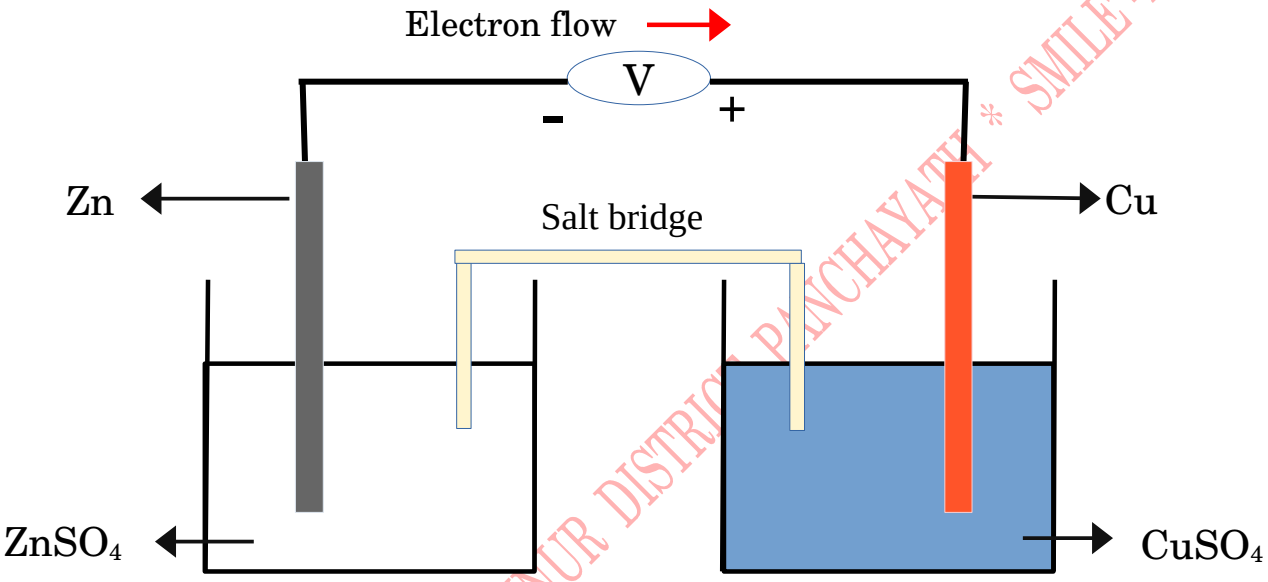
Galvanic cells are devices that convert chemical energy into electrical energy. Daniel cell is an example of a galvanic cell.

Two half cells are required to make a Daniel cell. A metal in its salt solution is a half cell. Zinc rod in the zinc sulphate solution taken in a beaker is one half cell and copper rod in the copper sulphate solution in another beaker is second half cell. Connect the solutions using a salt bridge. Potassium chloride or ammonium chloride or potassium nitrate etc. can be use to make salt bridge. Connect the zinc rod and copper rod to LED using copper wires. The LED glows.

Separate the copper wires from the bulb and connect them to a galvanometer. From the deflection it is clear that the flow of electrons is from the zinc rod to the copper rod and that of electric current is in the opposite direction.

Connect a voltmeter instead of the galvanometer to find the voltage between the two electrodes.

Diagram of a Daniel cell made by using Zn and Cu metal is given below.



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The reactions that occurs at each electrode is given below.

Zinc electrode (More reactive)	Copper electrode (Less reactive)
Negative electrode	Positive electrode
Oxidation take place at Zinc (Loosing of electron)	Reduction take place at Copper (Gaining of electron)
Zn electrode is anode	Cu electrode is cathode
$Zn \longrightarrow Zn^{2+} + 2e$	$Cu^{2+} + 2e \longrightarrow Cu$

The electrode where oxidation occurs is called the anode, and the electrode where reduction occurs is called the cathode.

The processes that takes place in this Daniel cell can be summarised as:

Anode : $Zn(s) \rightarrow Zn^{2+}(aq) + 2e-$

Cathode : $Cu^{2+}(aq) + 2e- \rightarrow Cu(s)$

Cell reaction :

$$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$$

PANCHAYATH * SMILE -2025

Reactivity series

The series in which elements are arranged in the decreasing order of their reactivity is known as reactivity series.

Potassium	K
Sodium	Na
Calcium	Ca
Magnesium	Mg
Aluminium	Al
Zinc	Zn
Iron	Fe
Nickel	Ni
Tin	Sn
Lead	Pb
Hydrogen	H
Copper	Cu
Mercury	Hg
Silver	Ag
Gold	Au

Reactivity decreases →

Complete the table

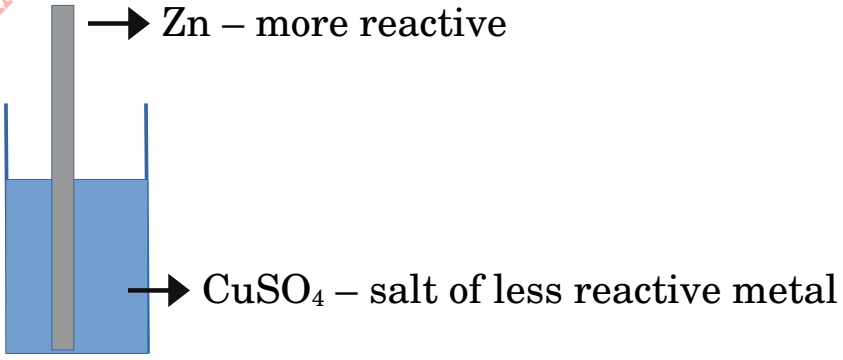
Electrodes used in Daniel cell	Anode	Cathode	Anode reaction	Cathode reaction	Cell reaction
Al - Cu					
Zn - Fe					
Ni - Cu					
Al - Ag					

Applications of the reactivity series

1. To identify which metal is being displaced.

The more reactive metal displace the less reactive metal from its salt solution.

Eg1: Zn is placed in copper sulphate solution



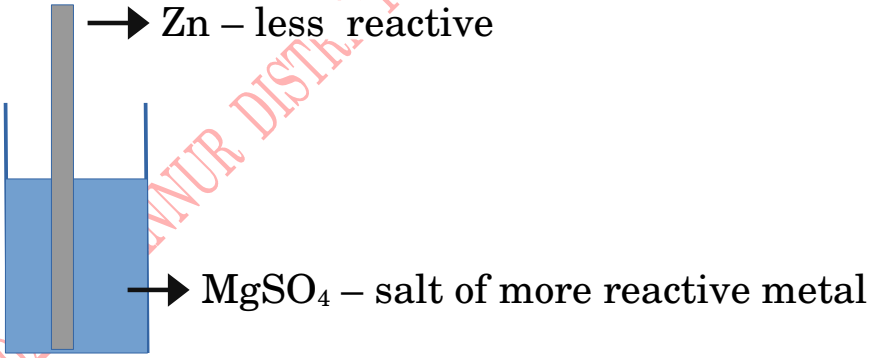
Here more reactive Zn metal is placed in the salt of less reactive metal Cu. So displacement reaction take place. Cu metal deposited over Zn. Blue colour of $CuSO_4$ fades.

$$Zn + CuSO_4 \longrightarrow Cu + ZnSO_4$$

More reactive metal underwent oxidation Here more reactive metal is Zn Zn underwent oxidation (Loosing of electron) $Zn \longrightarrow Zn^{2+} + 2e$	Less reactive metal is Cu Ion of less reactive metal underwent reduction Cu^{2+} underwent reduction (Gaining of electron) $Cu^{2+} + 2e \longrightarrow Cu$

A decrease in the concentration of Cu^{2+} ions is the reason for the fading of blue colour.

Eg 2 : Zn is placed in $MgSO_4$ solution



Here less reactive Zn metal is placed in the salt of more reactive metal Mg. Therefore, the displacement reaction will not take place.

2. To identify oxidising agent and reducing agent

The more reactive metal gets oxidised while the ion of the less reactive metal gets reduced.

The substance that causes oxidation is called an oxidising agent and the substance that causes reduction is called a reducing agent.

In a redox reaction, the oxidising agent gets reduced and the reducing agent gets oxidised.

In redox reactions, the more reactive metal acts as a reducing agent and the less reactive metal acts as an oxidising agent.

$$Mg^{0} + Cu^{2+}S^{6+}O^{2-}_{4} \rightarrow Mg^{2+}S^{6+}O^{2-}_{4} + Cu^{0}$$

More reactive metal is Mg

So Mg gets oxidised.

Mg is reducing agent

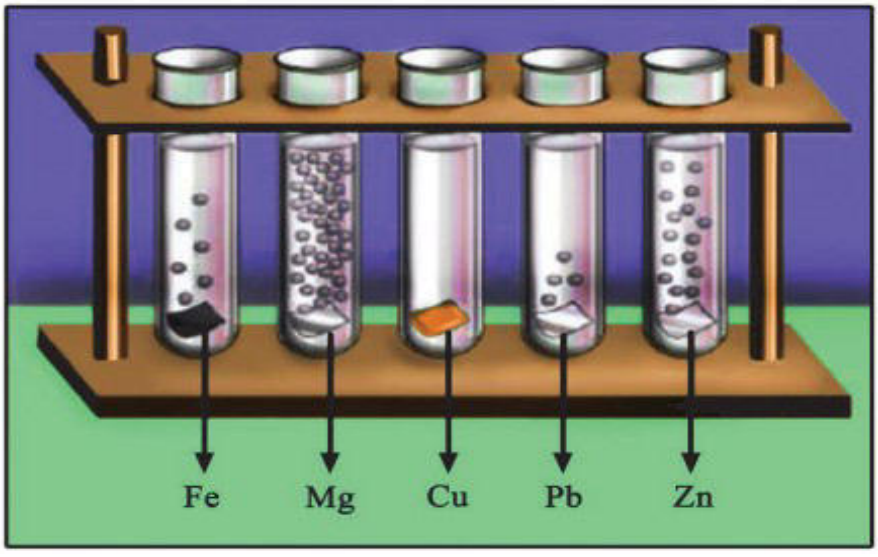
Less reactive metal is Cu

So Cu^{2+} ion gets reduced.

Cu^{2+} is oxidising agent

3. To identify the displacement of hydrogen from acid

The metals placed above hydrogen in the reactivity series can displace hydrogen from dilute acids.



Take equal amounts of dilute HCl in different test tubes as shown in the above Figure. Treat equal masses of polished Fe, Mg, Cu, Pb, Zn with dilute hydrochloric acid.

In which of the test tubes is hydrogen gas produced?

Is the rate of formation of hydrogen gas the same in all the test tubes?

Electrolytic cells

Electrolytic cells are the cells which utilise electrical energy to bring about chemical changes

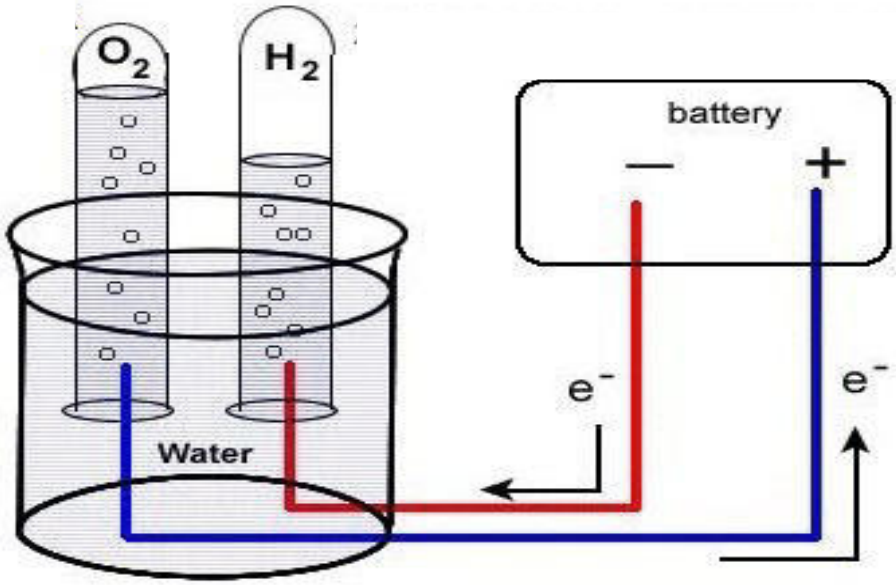
> Electrolytes are substances that conduct electricity in molten state or in an aqueous solution and undergo chemical changes. Electrolysis is the process by which an electrolyte undergoes chemical changes when electricity is passed through it.

Electrolysis of acidified water

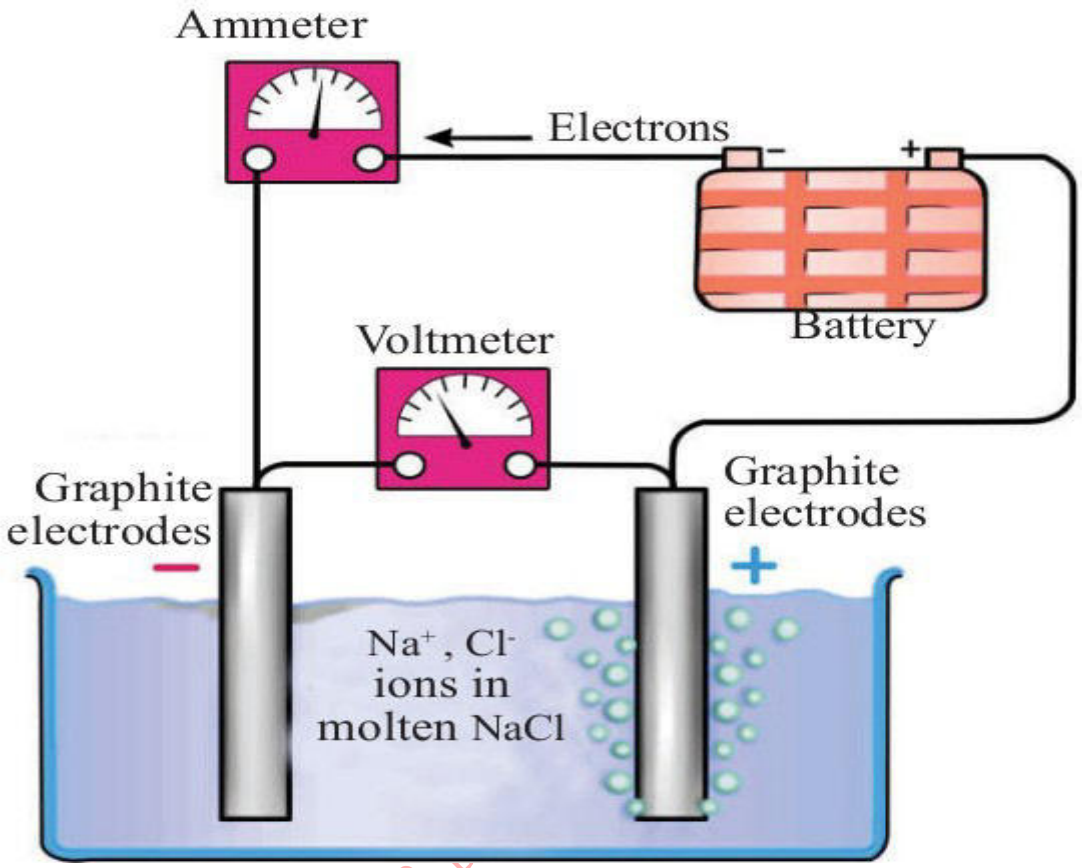
Acidified water decomposes into hydrogen and oxygen.

Oxygen gas (O_2) is liberated at the anode(positive electrode).

Hydrogen gas (H_2) is liberated at the cathode (negative electrode).



Electrolysis of molten sodium chloride



Reaction at cathode (negative electrode)

$$Na^+ + 1e \longrightarrow Na$$

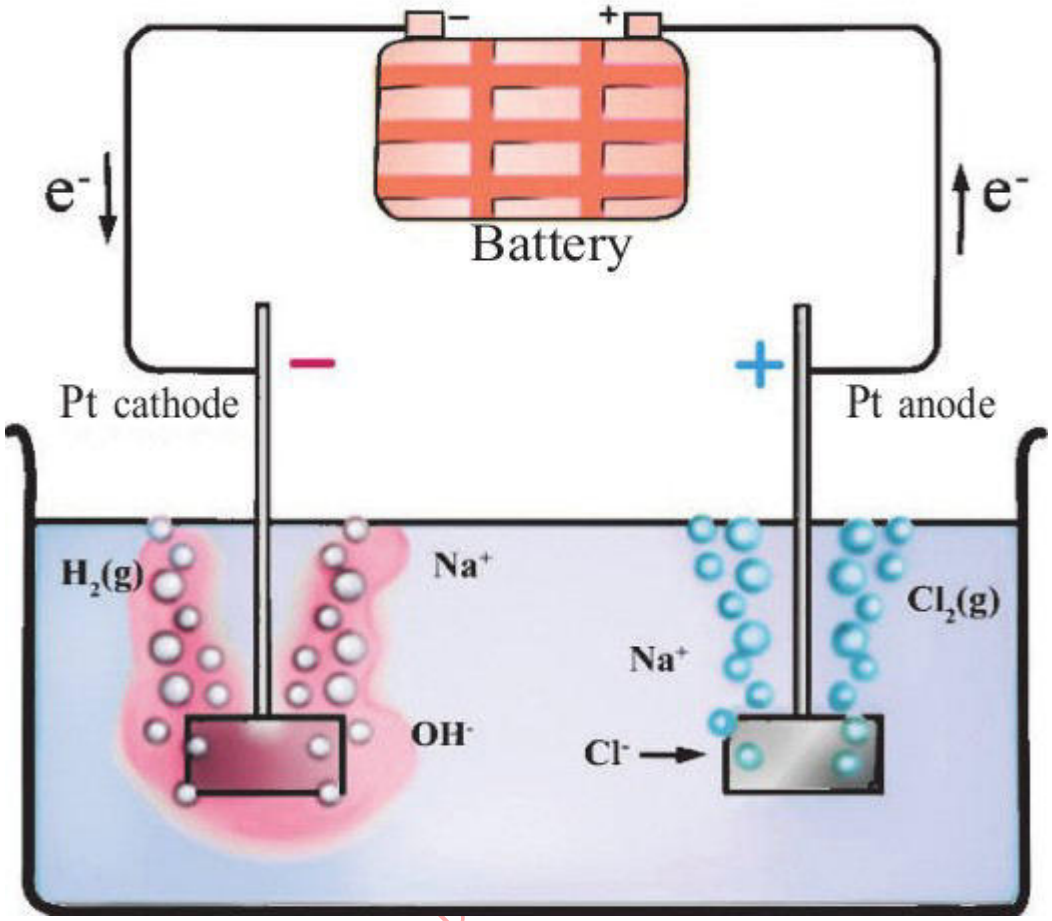
Silvery molten sodium metal (Na) is formed at the cathode.

Reaction at anode (positive electrode)

$$2Cl^- \longrightarrow Cl_2 + 2e$$

Chlorine gas (Cl_2), having a pale green colour, is liberated at the anode.

Electrolysis of aqueous sodium chloride solution



H_2 (hydrogen) gas is liberated at the cathode.

If phenolphthalein indicator is present in the solution, the solution around the cathode turns pink. That is, the solution becomes basic in nature. It is due to the formation of an aqueous solution of NaOH.

Cl_2 (Chlorine) gas is liberated at the anode.

Reaction at cathode (negative electrode)

$$2H_2O + 2e^- \longrightarrow H_2 + 2OH^-$$

Reaction at anode (positive electrode)

$$2Cl^- \longrightarrow Cl_2 + 2e$$

Electroplating

Electroplating is the process of coating a layer of one metal onto another metal through electrolysis.

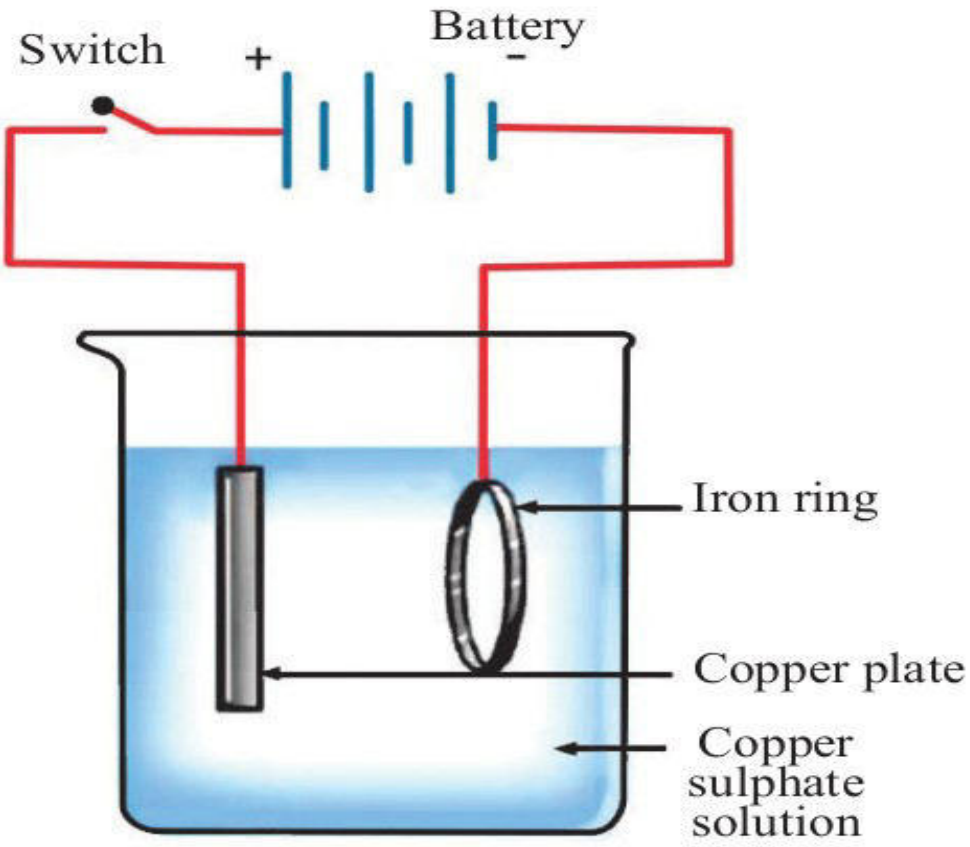
The coating thus formed not only enhances the beauty of the metal but also prevents its corrosion.

The object to be coated is connected to the negative terminal of the battery.

The metal which is used for plating is connected to the positive terminal of the battery.

A salt solution of the metal to be plated is used as the electrolyte.

STRICT PANCHAYATH * SMILE -2025



Reaction at anode (positive electrode)

$$Cu \longrightarrow Cu^{2+} + 2e$$

Reaction at cathode (negative electrode)

$$Cu^{2+} + 2e \longrightarrow Cu$$

That is, at the anode, copper undergoes oxidation and enters into the solution as Cu^{2+} ions. At the same time, Cu^{2+} ions from the solution are reduced on the surface of the iron ring, which is the cathode, forming a thin coating of copper on it.

The metal to be coated	Electrolyte
Silver	Silver nitrate solution OR sodium cyanide + silver cyanide solution
Gold	Sodium cyanide + gold cyanide solution
Copper	Copper sulphate solution

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Electrochemical cells as energy sources

A cell is a device that helps to convert the energy released in a chemical reaction directly into electricity. Cells perform two main functions.

1. As portable sources of electrical energy:
   button cells, dry cells, the lead acid cells used in vehicles.

2. As storage devices of electrical energy provided by an external source:
   cells can be used for powering electric vehicles,
   emergency power distribution and storing solar energy.

Primary cells

Cells cannot be recharged and reused.

Eg: dry cells , button cells.

Secondary cells

A secondary cell can be recharged and used again.

Eg: lead acid cell , The nickel-cadmium cell ,Lithium ion batteries

Fuel cells

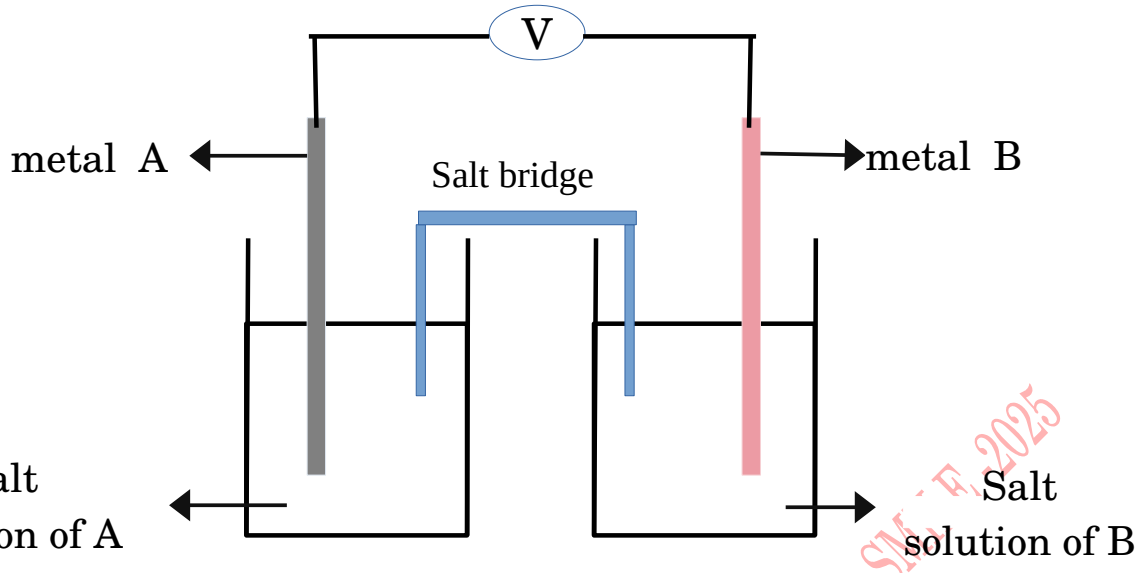
Fuel cells are galvanic cells designed to convert the chemical energy produced by the combustion of fuels such as hydrogen, methane and methanol into electrical energy. They produce electricity with high efficiency.

ATH * SMILE -2025

Applications of electrolysis

1.	Production of metals	: Metals like sodium, potassium, calcium and aluminium are produced by the electrolysis of some of their compounds.
2.	Production of nonmetals	: Electrolysis is used in the industrial production of nonmetals like hydrogen, oxygen and chlorine.
3.	Manufacture of compounds	: Electrolysis is used in the production of sodium hydroxide and potassium hydroxide.
4.	Electroplating	: Gold plated jewellery, silver plated utensils, chromium plated iron objects etc are produced through electroplating.
5.	Purification of metals	: Electrolysis is useful in the purification of metals such as copper, gold etc.

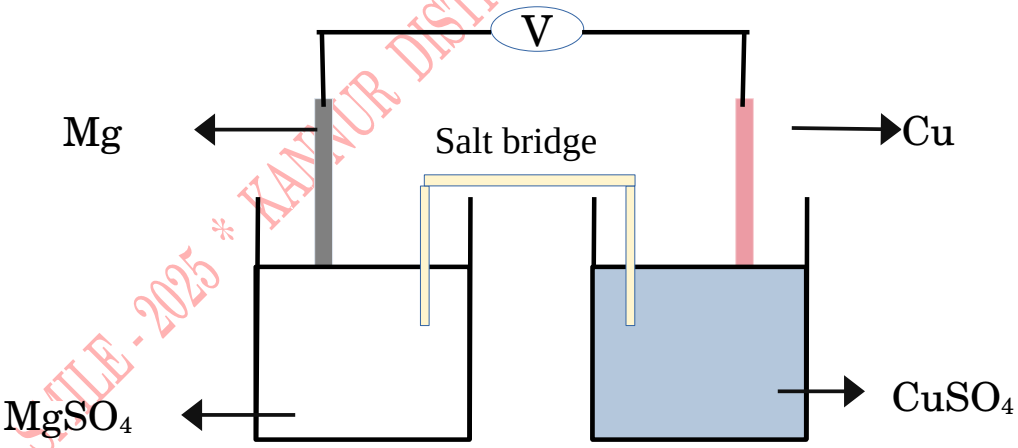
1. Diagram of a galvanic cell is given below.



Cell reaction $3A + 2B^{3+} \longrightarrow 3A^{2+} + 2B$

a) Which is anode in this cell ?
b) Write the equation take place at cathode.
c) Write equation of oxidation reaction..

2.

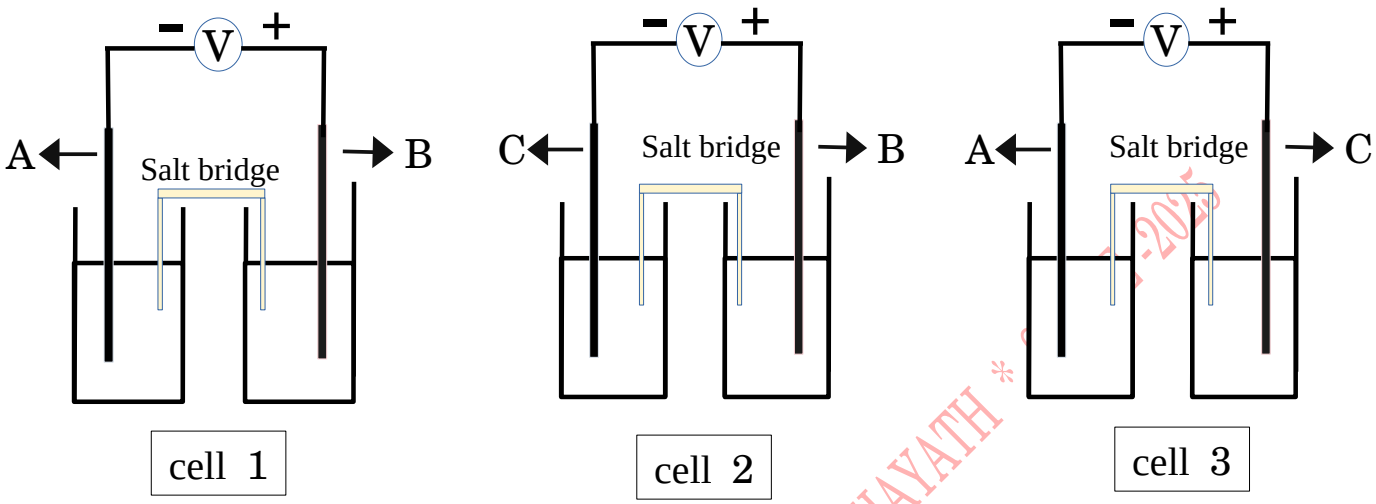


a) Which metal's mass decreases during a chemical reaction?
(Mg / Cu)
b) At Which electrode oxidation takes place?
c) In which of the following does the concentration of metal ions increase during a chemical reaction?
($MgSO_4$ / $CuSO_4$)
d) Write the cell equation.

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3. Galvanic cell: Chemical energy is converted into electrical energy.
…….: Electrical energy is converted into chemical energy.

4. The following are the pictures of 3 galvanic cells made using three metals A, B and C. (Valency of all three metals = 3)



a) Arrange A, B, and C in decreasing order of their reactivity

b) Which of them will give higher voltage?

c) Write reduction equation in the cell 3.

5. An iron nail is placed in $CuSO_4$. Choose from the given which chemical reactions are being performed here.

i) $Fe \longrightarrow Fe^{2+} + 2e$
ii) $Fe^{2+} + 2e \longrightarrow Fe$
iii) $Cu \longrightarrow Cu^{2+} + 2e$
iv) $Cu^{2+} + 2e \longrightarrow Cu$

a) (i) , (ii) reactions takes place
b) (i) , (iv) reactions takes place
c) (i) , (iii) reactions takes place
d) (ii) , (iii) reactions takes place

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6. Complete the table

Electrolyte	The resulting product		Chemical equation	
	Anode	Cathode	Anode	Cathode
Acidified water				
Molten NaCl				
Aqueous solution of NaCl				

7. The following are certain observations about metals A, B, C and D.

(a) When a plate of metal A is placed in a solution containing B^{2+} ions, no reaction is observed.
(b) When the plate of A is placed in a solution containing C+ ions, no change occurs.
(c) When a plate of metal D is placed in a solution containing C^{+} ions, C is formed on the surface of D .
(d) When a plate of B is placed in a solution of D^{2+} ions, D appears on the surface of B.

a) Arrange these metals in the increasing order of their reactivity.
b) Which metals are to be used to make a galvanic cell with the highest voltage?
c) Write the reaction take place at anode of galvanic cell made by using metals " C " and "D ".

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8. A galvanic cell is made up of the following half cells:
*A magnesium electrode immersed in magnesium sulphate solution.
* A nickel electrode immersed in nickel sulphate solution.

a) At which electrode oxidation takes place?
b) Select the direction of the electric current in this cell?
i) From the magnesium electrode to the nickel electrode
i) From the nickel electrode to the magnesium electrode
c) Write the cell equation.

6 METALS

The discovery of metals and their role in human progress is amazing. The term 'metal age' refers to the transition period from the stone age to the use of metals. Our ancient period is marked by the names copper age, bronze age, iron age etc. Metals are used in many forms in different fields of our life.

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Properties of Metals

1) Sonority
2) Malleability
3) Ductility
4) Thermal Conductivity
5) Hard
6) Electrical Conductivity
7) Metallic Lustre

Minerals

Naturally occurring metals or compounds are generally called minerals. These include metallic compounds and non metallic compounds.

For example, clay ($Al_2O_3.2SiO_2.2H_2O$) is an abundant mineral of aluminium. Bauxite ($Al_2O_3.2H_2O$) and cryolite (Na_3AlF_6) are also minerals of aluminium

The qualities of a mineral which is used to extract a metal

* Abundance
* Ease of extraction
* High metal content
* Low production cost.

Any mineral from which a metal can be extracted easily and economically is called an **ore of that metal.**

Metal	Ore	Chemical formula
Aluminium(Al)	Bauxite	$Al_2O_3.2H_2O$
Iron (Fe)	Haematite	Fe_2O_3
	Magnetite	Fe_3O_4
Copper (Cu)	Copper pyrites	$CuFeS_2$
	Copper glance	Cu_2S
	Cuprite	Cu_2O
Zinc (Zn)	Zinc blende	ZnS
	Calamine	$ZnCO_3$
Tin (Sn)	Cassiterite / Tinstone	SnO_2

Metallurgy

The process of extraction of metal from ores involves three major stages. Such processes are generally referred to as metallurgy.

Concentration of ores	Extraction of metal from concentrated ore	Refining of metal

Concentration of ores

Ores are mined from the earth's crust. A lot of earthy impurities like soil and sand may be mixed with this. Concentration of ores is the process of increasing the metal content of the ore by removing such impurities, known as gangue. Depending on the nature of the ore and gangue, various methods are adopted for this purpose.

1. Levigation or Hydraulic washing

This method is employed when the density of the ore is more than that of the gangue.

Generally, oxide ores can be concentrated by this method

2.Froth floatation

The froth floatation process is used when the ore is less dense than the gangue.

Generally, sulphide ores are concentrated by this method.

3. Magnetic separation

When the powdered ore is passed through a magnetic roller, either the ore or the impurities having magnetic property is separated.

For example,

i) Magnetite (Fe_3O_4) ore has magnetic properties, but the gangue is non magnetic.

ii) Tinstone (SnO_2) ore is non magnetic, but iron tungstate, the gangue present in it is magnetic.

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4. Leaching

The powdered ore is mixed with a suitable solvent. The ore dissolves in it or undergoes a chemical reaction to become a solution. The gangue remains insoluble in it. It is then filtered out. This process is called leaching.

Eg:

i) Bauxite, the ore of aluminium, is concentrated in this way using NaOH as a solvent

ii) Metals such as gold and silver occur almost freely in nature. The ores of these metals are leached with a dilute solution of sodium cyanide (NaCN) or potassium cyanide (KCN) in the presence of air to extract the metals.

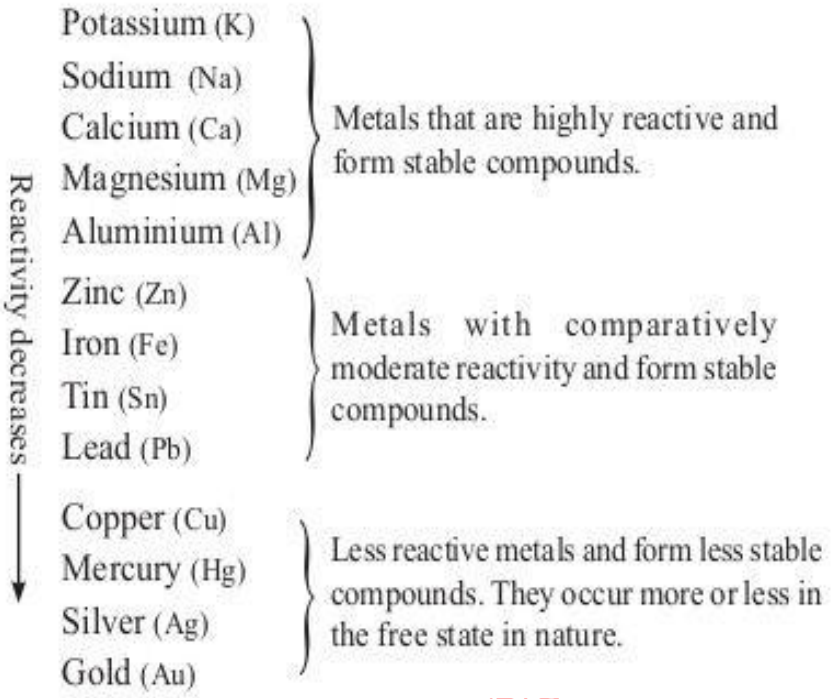
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The nature of ores is given. Complete the table using appropriate concentration methods.

Nature of ore	Concentration method
1. Ores are less dense than the impurities.	.....
2. Ores have magnetic properties but impurities are non magnetic.	.....
3. A solvent which dissolves the ore is used.	.....
4. Ores are denser than the impurities.	.....

Extraction of metal from concentrated ore

metals in the reactivity series.



Mercury can be easily separated from its ores by controlled heating.

Metals with comparatively moderate reactivity like Zn, Fe, Sn, Pb, are generally found as oxides, sulphides or carbonates.
Eg. $ZnCO_3$, ZnS, Fe_2O_3, SnO_2 , PbS.

There are two stages for the production of metal from them.

1. Conversion of concentrated ore to oxide

Two methods are commonly used for this.

(a) Calcination

Metal carbonates or hydroxides are converted to their oxides by a process called calcination. Calcination refers to heating the concentrated ore in limited quantities or absence of air at a temperature below the melting point.

Eg. $CaCO_3(s) \xrightarrow{heat} CaO(s) + CO_2(g)$

$ZnCO_3(s) \xrightarrow{heat} \ldots\ldots\ldots + \ldots\ldots\ldots$

b) Roasting

Roasting refers to heating the concentrated ore at a temperature below the melting point in the presence of air. Zinc sulphide and lead sulphide can be easily converted into their oxides by this method.

$$2ZnS(s) + 3O_2(g) \xrightarrow{heat} 2ZnO(s) + 2SO_2(g)$$

$$2PbS(s) + \ldots\ldots\ldots\ldots \xrightarrow{heat} \ldots\ldots\ldots\ldots + \ldots\ldots\ldots\ldots$$

When the concentrated ore is subjected to roasting, the water content in them gets released as vapours. They become pure and moisture free oxide ores.

2. Reduction of oxide ore

The metal part of the ores is in the form of positive ions, and hence it can be extracted by a reduction process. Reducing agents such as coke (an allotrope of carbon) and carbon monoxide are used to extract moderately reactive metals such as iron, zinc and tin from their oxide ores. Carbon has a greater tendency to react with oxygen at high temperatures, so oxygen is easily removed from the ore. Strong reducing agents are required to extract highly reactive metals such as sodium, potassium, calcium, magnesium and aluminium from their ores. Such metals are produced by electrolysis.

Method of separating metals from ores	Metals
Electrolysis	K , Na, Ca, Mg, Al
Reduction using coke	Zn
Heating in controlled ways	Hg
Reduction using carbon monoxide	Fe

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Refining of metals

The metal extracted from concentrated ore may contain other elements or their compounds as impurities. Refining of metals is the process of removal of these impurities to produce pure metal.

Various methods are adopted for this purpose depending upon the nature of the metal to be purified and the impurities contained in it. Some are given below.

(a) Liquation

When metals with low melting point contain impurities with high melting point, the metal is heated on the inclined surface of the furnace. The pure metal separates from the impurities, melts, flows down and it is collected. This is known as liquation.

Metals such as tin (Sn) and lead (Pb) having low melting point can be purified by this method.

b) Distillation

This method is used to purify metals with relatively low boiling points. When the metal containing impurities is heated under suitable conditions, only the pure metal vapourises. When this vapour condenses, pure metal is obtained.

Metals like zinc (Zn), cadmium (Cd) and mercury (Hg) can be purified using this method.

(c) Electrolytic refining

In an electrolytic cell, an impure metal is used as the anode and a thin piece of pure metal is used as the cathode. A suitable salt solution of the same metal is used as the electrolyte. When electric current is passed, the pure metal alone is separated from the anode and deposited at the cathode.
Copper (Cu) and zinc (Zn) can be purified in this way.

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Extraction of iron

Iron comprises about 5% of the earth's crust and it is the second most abundant metal.

Iron is produced in steel towers, known as blast furnaces, The concentrated ore (Fe_2O_3) mixed with limestone ($CaCO_3$) and coke (C) is fed from the top of the furnace via cup & cone arrangement. This mixture is called charge.

A blast of hot air is blown into the furnace from its bottom. . The coke combines with the oxygen in the air, when the blast of hot air flows from the bottom to the top of the blast furnace.

$$C + O_2 \rightarrow CO_2 + \text{heat}$$ (exothermic reaction.)

Coke reacts with carbon dioxide to produce carbon monoxide. This is an endothermic reaction.

$$CO_2 + C + \text{heat} \rightarrow 2CO$$

The main chemical reaction taking place in the blast furnace is the reduction of iron ore using carbon monoxide.

$$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$$

In the production of iron, limestone is added to the ore as a source of flux . The flux combines with gangue to form molten slag.

The limestone decomposes to calcium oxide (Quick lime). Calcium oxide produced in this reaction acts as flux.

$$CaCO_3\,(s) \xrightarrow{\text{heat}} CaO(s) + CO_2\,(g)$$

Calcium oxide reacts with sand (SiO2) to form calcium silicate (slag) which melts easily.

$$CaO(s) + SiO_2(s) \rightarrow CaSiO_3(l)$$

Flux Gangue Slag

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If the gangue in the ore is basic in nature, acidic flux should be used and If the gangue in the ore is acidic in nature, basic flux should be used

Pig iron

The molten iron obtained from the blast furnace contains 4% carbon and small amounts of other impurities such as manganese, silicon, phosphorus and sulphur. This is called pig iron.

Cast iron

If the percentage of impurity in it is reduced, the iron obtained can be cast as desired. It is called cast iron. It contains more than 2% carbon.

Wrought iron

Wrought iron is a relatively pure form of iron, containing 0.02% - 0.05% carbon and small amounts of phosphorus, silicon, manganese, sulphur etc.

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Steel

Carbon steel	Amount of carbon (%)	Features
Mild steel	0.05 - 0.2	Easy to draw into wires and hammer into plates. Strong and hard. Used for making agricultural tools.
Medium steel	0.2 - 0.6	High hardness. Used for the construction of railway tracks, handrails, rafters etc.
High carbon steel	0.6 - 1.5	Very high elasticity and hardness. Used for the manufacture of surgical instruments, springs, knives, drills etc.

Alloy steel

Alloy steels	Constituent elements	Properties and uses
Stainless steel	Fe, Cr, Ni, C	Resists corrosion. Manufacture of household appliances and blades.
Manganese steel	Fe, Mn, C	High hardness. Used for the manufacture of drills, safes, plates and railway tracks.
Silicon steel	Fe, Si, C	Also known as electrical steel. As it reduces electric loss, it is used in the core of electromagnetic instruments like motors, generators, transformers etc.

Extraction of zinc

Zinc is another industrially important metal.

Sulphide ore of zinc (Common name, chemical formula)	.....
Concentration method	.....
Carbonate ore of zinc (Common name, chemical formula)	.....

Zinc has comparatively low boiling point (907°C) and hence it vapourises at high temperature and comes out. It is immediately cooled to liquid form.

$$ZnO(s) + C(s) \xrightarrow{\text{heat}} Zn(s) + CO(g)$$

Extraction of copper

Principal ores of copper

Copper pyrites – $CuFeS_2$

Copper glance – Cu_2S

Sand (SiO_2) having acidic nature(flux) is added to the ore and heated to form cuprous sulphide (Cu_2S) of maximum purity. It is partially converted to cuprous oxide (Cu_2O) when heated under controlled flow of air. The cuprous oxide converts the remaining cuprous sulphide into copper. No reducing agent is added here.

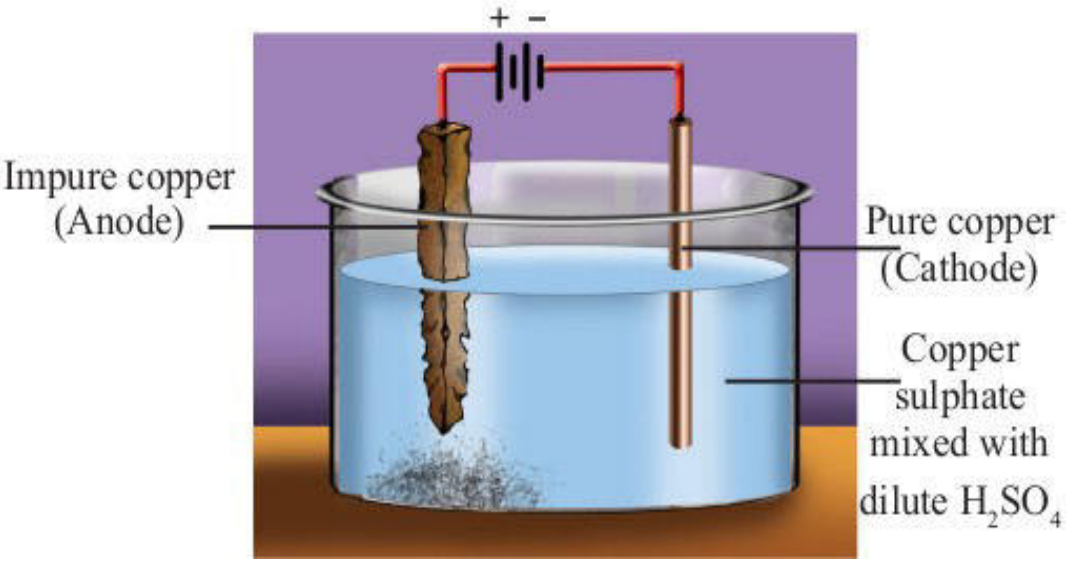
$$2Cu_2O + Cu_2S \rightarrow 6Cu + SO_2\uparrow$$

Molten copper is cooled in moulds. The copper lumps thus obtained are called blister copper. The sulphur dioxide gas produced during the production of copper is released when the metal condenses. As a result blisters are formed and they appear as such on the surface of copper lumps. That is why it is called blister copper

Brass and bronze are important alloys that contain copper.

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Purification of Copper



Anode	
Cathode	
Electrolyte	
Chemical equation of the reaction taking place at anode.	
Chemical equation of the reaction taking place at cathode.	

Extraction of aluminium

Aluminium is produced by Hall-Heroult process. It has two stages.

1. Concentration of bauxite

Bauxite is concentrated by leaching. The powdered ore is treated with hot concentrated sodium hydroxide (NaOH) solution.

The ore alone dissolves in it to form sodium aluminate ($NaAlO_2$) solution. Impurities are then filtered off.

$$Al_2O_3.2H_2O + 2NaOH \rightarrow 2NaAlO_2 + 3H_2O$$

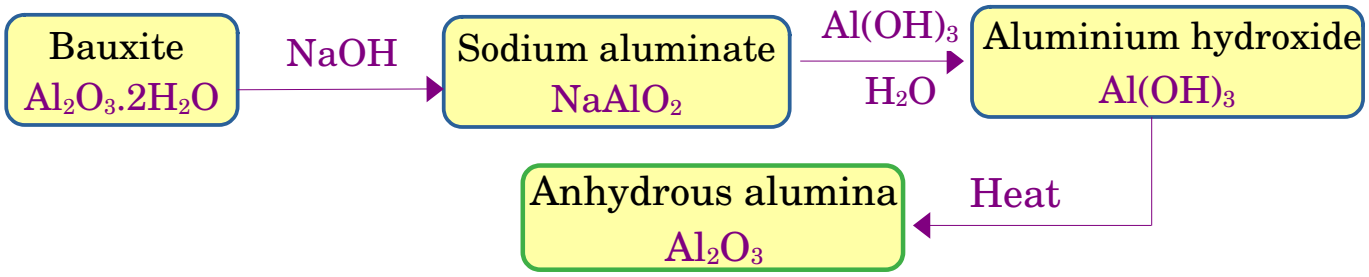
A little aluminium hydroxide ($Al(OH)_3$) is added to the solution of sodium aluminate and stirred vigorously. It is diluted by adding water. As a result, pure aluminium hydroxide separates out from the solution.

$$NaAlO_2 + 2H_2O \rightarrow Al(OH)_3 + NaOH$$

If the aluminium hydroxide thus obtained is heated strongly, anhydrous alumina (Al_2O_3) is obtained.

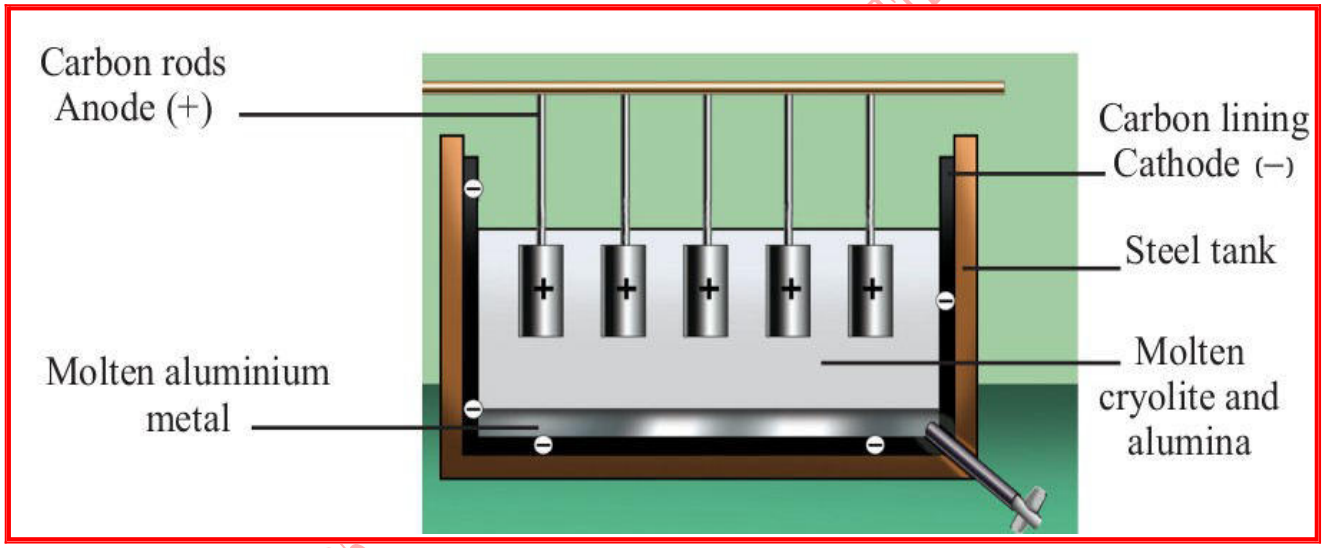
$$2Al(OH)_3 \xrightarrow{heat} Al_2O_3 + 3H_2O$$

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2. Electrolysis of aluminium oxide

Molten cryolite (Na_3AlF_6) is added to the alumina, is the electrolyte used in the production of aluminium. The melting point of alumina is 2017°C. Cryolite is added to reduce the melting point and to increase the electrical conductivity



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Aluminium is separated at the cathode when a high voltage electric current is passed through the molten mixture. Examine the chemical equations of the reactions.

$$Al_2O_3 \rightarrow 2Al^{3+} + 3O^{2-}$$

At the cathode

$$Al^{3+} + 3e^- \rightarrow Al$$

At the anode

$$2O^{2-} \rightarrow O2(g) + 4e^-$$

Complete the table related to electrolysis of aluminium.

Anode	
Cathode	
Electrolyte	
The product formed at anode	
The product formed at cathode	
Reducing agent	

Corrosion of metals

Corrosion of metal is a process in which the metal reacts with a surrounding medium and undergoes chemical change.

The factors given below influence the corrosion of metals.

* Nature of the metal
* Moisture in which gases or salts are dissolved
* Contact with other metals

- When a metal comes in contact with another metal in the presence of an electrolyte, it acts as a simple voltaic cell.

- When a metal is in contact with another metal in the presence of moisture in the air,The more reactive metal undergoes Corrosion.

It is seen that the metal acting as anode is destroyed(more reactive metal) and the metal acting as cathode is protected(less reactive metal). This is an important method to prevent corrosion of iron. This is known as cathodic protection.

Eg: Magnesium blocks always attached to sea bridges and ship hulls to prevent corrosion

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7 SOME COMPOUNDS OF INDUSTRIAL IMPORTANCE

Agriculture is something that man has learnt from nature. Ours is a land of agricultural abundance. The country has been able to achieve self-sufficiency in food through improved farming methods. Chemistry plays a major role in this.

Ammonia (NH_3)

Ammonia is a chemical used for the production of nitrogenous fertilizers which are required for the growth of plants.

Preparation of Ammonia

Take a little ammonium chloride (NH_4Cl) in a watch glass. Add a little calcium hydroxide ($Ca(OH)_2$) to it and stir well using a glass rod.

$$2NH_4Cl + Ca(OH)_2 \rightarrow \text{.....} + \text{.....} + \text{.....}$$

Properties of Ammonia

- Pungent smell
- Basic nature
- Red litmus paper -Blue
- Ammonia is water soluble
- Density of ammonia is less than that of air

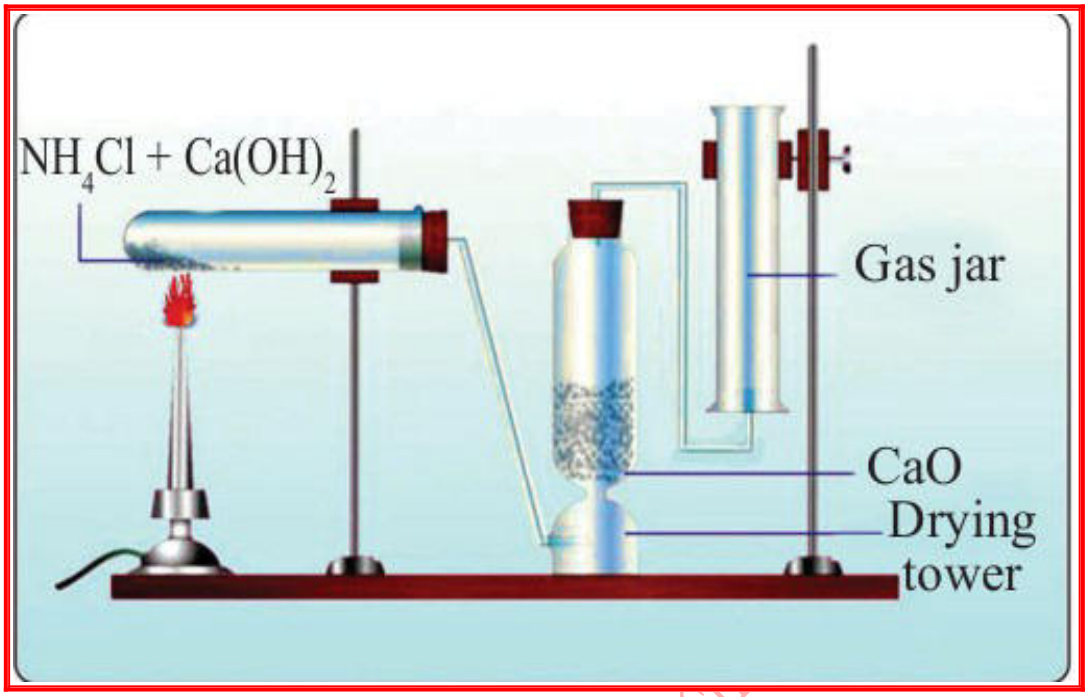
$$NH_3 + H_2O \rightarrow NH_4OH$$

The preparation of ammonia in the laboratory

Ammonia is prepared in the laboratory by heating the mixture of ammonium chloride (NH_4Cl) and calcium hydroxide ($Ca(OH)_2$).

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When ammonia is passed through quick lime (CaO) in the drying tower, the moisture contained in it is removed.



In Figure, the jar used for collecting ammonia is kept inverted, why?

Industrial preparation of ammonia (Haber process)

In this process, nitrogen and hydrogen, in the ratio 1:3, are made to combine at very high pressure and temperature to produce ammonia. Sponge iron is used as the catalyst.

$$N_2(g) + 3H_2(g) \; 2NH_3(g) + heat$$

Atmospheric air is the source of nitrogen. Petroleum products like naphtha are used as sources of hydrogen. Hydrogen obtained by the electrolysis of water is also used.

The ammonia produced by Haber process is liquefied under high pressure and stored as liquid ammonia.

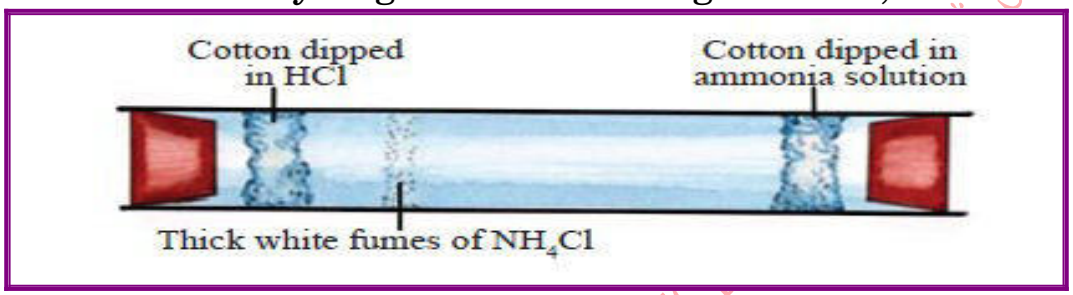
Liquor ammonia is the concentrated aqueous solution of ammonia. Ammonia gas can be easily liquefied by applying pressure. Liquefied ammonia is known as liquid ammonia.

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Uses of ammonia

- For the preparation of chemical fertilizers such as ammonium sulphate, ammonium phosphate and urea.
- As refrigerant.
- For the production of artificial fibres such as nylon and rayon.
- In the refining of petroleum.
- As the main raw material in the industrial preparation of nitric acid by Ostwald process

Let us do an experiment showing the chemical reaction between ammonia and hydrogen chloride in a glass tube, as shown in the picture.



Place cotton dipped in HCl at one end and another piece of cotton dipped in ammonia solution at the other end of the glass tube, such that these are well inside the glass tube. Close both ends of the glass tube using cork. Observe the changes inside the glass tube.

When the glass tube is heated, the ammonium chloride decomposes.

Write the chemical equation of the formation of ammonium chloride.

.....

Write the chemical equation of the decomposition of ammonium chloride.

.....

Let us write these chemical equations together.

$$NH_3 + HCl \rightleftharpoons NH_4Cl$$

The sign $\rightleftharpoons$ indicates that the reaction takes place in both directions.

Reactions taking place in both directions simultaneously are called reversible reactions.

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Examine the reversible reactions given bellow and write the forward and backword reactions.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

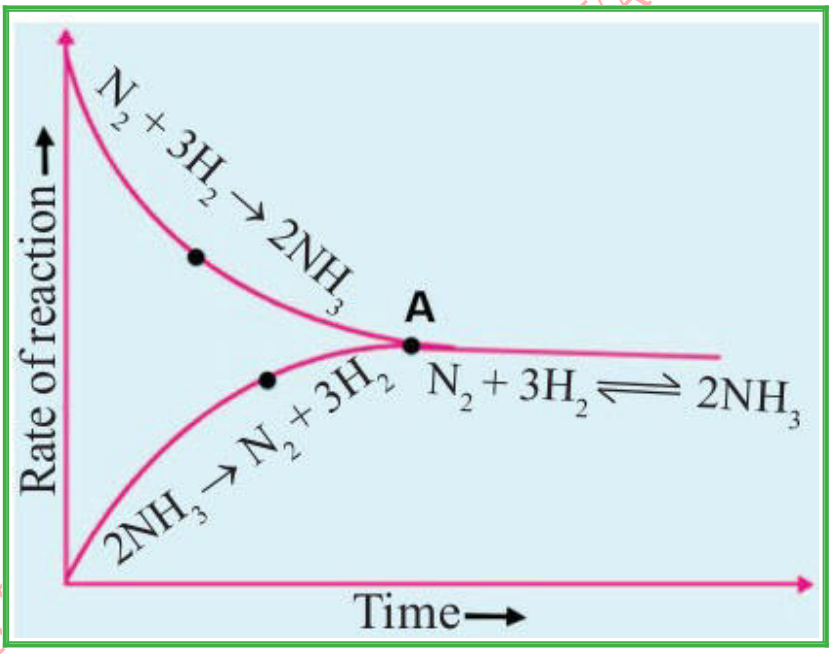
Irreversible reactions

$$NaOH + HCl \rightarrow NaCl + H_2O$$

$$C + O_2 \rightarrow CO_2$$

In all these reactions, reactants react to give products. However, at the same conditions, the products do not change to reactants. Such type of reactions is called irreversible reactions.

The formation of ammonia by the reaction of nitrogen and hydrogen is a reversible reaction.



A graph related to rate of reaction is given in the above figure. Analyse the graph and answer the questions given.

Only the reactants are present at the beginning of the reaction. Then, in which direction does the reaction progress?

Forward reaction / backward reaction

What happens to the rate of the forward reaction as time proceeds?

.....

What change occurs in the concentration of the product, as the chemical reaction proceeds?

.....

Then, what happens to the rate of backward reaction?

.....

Identify the point at which the rates of forward and backward reactions become equal?

.....

The stage at which the rate of forward reaction becomes equal to the rate of backward reaction in a reversible reaction is called chemical equilibrium.

Le Chatelier principle

"When the concentration, pressure or temperature of a system at equilibrium is changed, the system will readjust itself so as to nullify the effect of that change and attain a new state of equilibrium." This is Le Chatelier's principle.

Characteristics of equilibrium

- At equilibrium, both the reactants and the products coexist.
- The rates of forward and backward reactions are equal at equilibrium.
- Even after attaining equilibrium, the reactant molecules react to form product molecules and, at the same rate, the product molecules react to form reactant molecules. Hence, chemical equilibrium is a dynamic equilibrium at the molecular level.
- When nothing new is added to a system and nothing is removed from the system, such a system is called a closed system. Equilibrium is possible only in a closed system.

Influence of concentration on equilibrium

Examine the chemical equation given below.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

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If the concentration of one of the reactants, nitrogen, is increased in this system at equilibrium, what changes are made by the system to rearrange itself according to Le Chatelier's principle? Tick (√) only the correct ones in the Table

More ammonia will be formed	✓
More hydrogen will be used	
Rate of forward reaction will increase	
Rate of backward reaction will increase	

Analyse the effect of change in concentration in the above equilibrium system and complete Table given bellow.

Action	**Change in concentration**	**Change in rate**
More hydrogen is added	Increases the concentration of reactant	Rate of forward reaction .....
More ..... is added	Increases the concentration of product	Rate of backward reaction increases
Ammonia is removed	Concentration of product	Rate of forward reaction increases
More nitrogen is added	Concentration of reactant increases	Rate of forward reaction .....
Nitrogen is removed	Concentration of reactant decreases	Rate of forward reaction .....
Hydrogen is removed	Concentration of reactant	Rate of backward reaction

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According to Le Chatelier principle, if the concentration of reactants is increased in a system at equilibrium, the rate of forward reaction increases. If the concentration of product is increased, the rate of backward reaction increases.

Examine the chemical equation of the formation of ammonia.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

Observe the above chemical equation and complete Table

How many moles of reactant molecules are present?	....
How many moles of product molecules are present?	....
Change that takes place in forward reaction	..... moles of reactant molecules react to form ..... moles of product molecules.
Change that takes place in backward reaction	..... moles of product molecules decompose to form …… moles of reactant molecules.
Change in volume in forward reaction	Volume ……. (decreases / increases)
Change in volume in backward reaction	Volume ……. (decreases / increases)

At fixed volume, what happens to the pressure of a gaseous system if the number of molecules is increased?

Increases / Decreases .....

According to Le Chatelier principle, when the pressure of a system at equilibrium is increased, the system will try to attain equilibrium by reducing the pressure. This is achieved by increasing the reaction in the direction where number of moles of molecules decreases.

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Identify the influence of pressure in the production of ammonia and complete Table

Number of molecules (at the beginning of chemical reaction)	**Action**	**Change in rate of chemical reaction**
Total number of reactant molecules ..... (more / less)	Increase the pressure	Rate of forward reaction ..... (Increases / Decreases) Rate of backward reaction ..... (Increases / Decreases)
Total number of product molecules ..... (more / less)	Reduce the pressure	Rate of forward reaction ..... (Increases / Decreases) Rate of backward reaction ..... (Increases / Decreases)

Examine the following chemical equation.

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

What is the total number of reactant molecules?

What is the total number of product molecules?

What is the relation between the total number of reactant molecules and the total number of product molecules?

Does pressure influence this reaction?

In a reversible reaction, if there is no change in the number of molecules of reactants and products in gaseous state, pressure will have no effect on the chemical equilibrium.

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Temperature and equilibrium

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) + \text{heat}$$

Here, which is the endothermic reaction?

Forward reaction / Backward reaction

On increasing the temperature, the system tries to reduce it by increasing the rate of endothermic reaction. As a result, the rate of backward reaction increases. So, temperature has to be reduced in order to increase the rate of forward reaction. But at low temperature, the number of molecules having threshold energy will be less. Therefore, the rates of forward and backward reactions get significantly reduced and hence the system will take more time to attain equilibrium. So, in the manufacture of ammonia, 450oC is taken as the optimum temperature.

In a system at equilibrium, increase in temperature will increase the rate of endothermic reaction. Decrease in temperature will increase the rate of exothermic reaction.

Catalyst and equilibrium

You know that catalysts are substances which increase the rate of reactions. In a reversible reaction, catalysts increase the rate of both forward and backward reactions to the same extent. As a result, the system reaches equilibrium very fast.

What is the catalyst used in the manufacture of ammonia? .....

Sulphuric acid (H_2SO_4)

Sulphuric acid is often called the 'King of Chemicals' because it is an industrially important chemical.

Sulphuric acid is manufactured by Contact process.

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Important steps in Contact process

Molten sulphur is burnt in purified and moisture free atmospheric air and converted to sulphur dioxide.

$$S(s) + O_2(g) \rightarrow SO_2(g)$$

Sulphur dioxide formed in the first step combines with oxygen, in presence of the catalyst vanadium pentoxide (V2O5), to form sulphur trioxide.

$$2SO_2(g) + O_2(g) \underset{}{\overset{V_2O_5}{\rightleftharpoons}} 2SO_3(g)$$

SO_3 is dissolved in concentrated sulphuric acid to produce oleum ($H_2S_2O_7$).

$$SO_3(g) + H_2SO_4(l) \rightarrow H_2S2O_7(l)$$

The oleum thus obtained is dissolved in water to produce sulphuric acid.

$$H_2S2O_7(l) + H_2O(l) \rightarrow 2H_2SO_4(l)$$

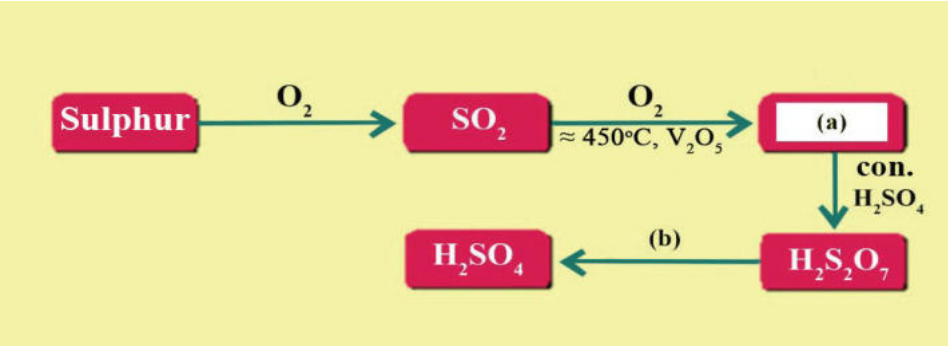
Complete the chemical equation given below to find out the product formed when sulphur trioxide dissolves in water.

$$SO_3 + H_2O \rightarrow .....$$

Dissolving sulphur trioxide in water gives sulphuric acid. Yet, why is sulphur trioxide not dissolved directly in water in the industrial production of sulphuric acid?

Since the dissolution of SO3 in water is an exothermic process, the sulphuric acid initially formed turns into fog like particles (smog) which will hinder further dissolution.

The flow chart of industrial production of sulphuric acid is given. Identify (a) and (b) in this and write it down.



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Physical properties of sulphuric acid

- Colourless
- Viscosity is higher than that of water.
- Denser than water.
- Dissolves in water.

Solubility in water

Take 5 ml water in a test tube and slowly add a few drops of concentrated sulphuric acid to it. Touch the bottom of the test tube. What do you feel?

Is the reaction exothermic or endothermic?

While diluting sulphuric acid, the acid has to be added to water in very small quantities by stirring. If dilution is done by adding water to acid, the chance of splashing out of acid leading to accident is very high.

Chemical properties of sulphuric acid

Dehydrating property

Take a little sugar in a watch glass and add a few drops of concentrated sulphuric acid. Observe the changes. Analyse the chemical equation of this experiment.

$$C_{12}H_{22}O_{11} \xrightarrow{con.H_2SO_4} 12C + 11H_2O$$

What is the black substance that remains? .....

What is the ratio of number of atoms of hydrogen and oxygen in a molecule of sugar? .....

What is the role of sulphuric acid in the chemical reaction?

Sulphuric acid absorbed the elements hydrogen and oxygen from sugar in the same ratio as in water.

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Dehydration is the process of absorbing chemically combined water, or hydrogen and oxygen in the same ratio as in water from substances. Concentrated sulphuric acid is a strong dehydrating agent.

Perform the following activities which demonstrate the dehydrating properties of sulphuric acid and record your observations.

Activity	**Observations**
Add a little con. sulphuric acid to glucose taken in a beaker.	.....
Add con. sulphuric acid drop by drop to blue copper sulphate taken in a watch glass.	.....

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Drying property

Drying agents are substances capable of absorbing the moisture present in a substance.

Concentrated sulphuric acid is used as a drying agent in the preparation of CO_2, SO_2 and HCl.

Concentrated H_2SO_4 is not used as a drying agent in the preparation of NH_3. Why? .....

Reaction with salts

Concentrated sulphuric acid reacts with chlorides to form hydrogen chloride and with nitrates to form nitric acid. Chemical equations of these reactions are given.

$$NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$$

$$KNO_3 + H_2SO_4 \rightarrow KHSO_4 + HNO_3$$

Concentrated sulphuric acid can displace acids from their salts.

This method is employed in the preparation of hydrochloric acid, nitric acid etc.

Oxidising nature

Analyse the following chemical equation.

$$C^{0} + 2H_2^{1+}S^{6+}O_4^{2-} \rightarrow C^{4+}O_2^{2-} + 2S^{4+}O_2^{2-} + 2H_2^{1+}O^{2-}$$

What is the oxidation state of elemental carbon? .....

What is the oxidation state of carbon in carbon dioxide?

.....

What happened to carbon in this reaction?

Oxidised / Reduced

Have a look at the chemical equation of the reaction between concentrated sulphuric acid and copper.

$$Cu^{0} + 2H_2^{1+}S^{6+}O_4^{2-} \rightarrow Cu^{2-}S^{6-}O_4^{2-} + S^{4+}O_2^{2-} + 2H_2^{1+}O^{2-}$$

Here, what change has occurred to copper? Examine by writing the oxidation state.

Oxidised / Reduced

Concentrated sulphuric acid reacts with metals as well as nonmetals and oxidise them. Concentrated sulphuric acid is a good oxidising agent.

Basicity of acids

You know that the presence of H+ ions is the basis for the characteristic properties of acids. Complete the following table by writing the equation of ionisation.

Acid	Equation of ionisation
HNO_3	$HNO_3 \rightarrow H^+ + NO_3^-$
HCl	.....
HBr	.....
HF	.....

How many hydrogen ions are liberated when each of the acid in the table ionises?

The basicity of an acid is the number of hydrogen ions that the molecule of the acid can donate. If basicity is 1, the acid is called monobasic acid.

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Examine the equation of ionisation of sulphuric acid.

$$H_2SO_4 \rightarrow H^+ + HSO_4^-$$

$$HSO_4^- \rightarrow H^+ + SO4^{2-}$$

If the basicity of an acid is 2, it is called dibasic acid.

Complete the following table by writing the equation of ionisation.

Acid	**Equation of ionisation**
H_2C2O_4 (HOOC-COOH)	$H_2C_2O_4 \rightarrow$ ..... + ...
H_2CO_3	$H_2CO_3 \rightarrow$ ..... + ...

Complete the equation of ionisation of phosphoric acid.

$$H_3PO_4 \rightarrow \text{.....} + PO4^{3-}$$

What will be the basicity of H_3PO_4 .....

If the basicity of an acid is 3, it is called tribasic acid.

Examine the equation of the reaction that takes place when sodium hydroxide dissolves in water.

$$NaOH \rightarrow Na^+ + OH^-$$

Complete the given table

Chemical name of alkali	**Chemical formula**	**Equation of ionisation in water**
Sodium hydroxide	NaOH	$NaOH \rightarrow Na^+ + OH^-$
Calcium hydroxide	.....	.....
Ammonium hydroxide	.....	.....

What is the common ion liberated when alkalise dissolve in water?

.....

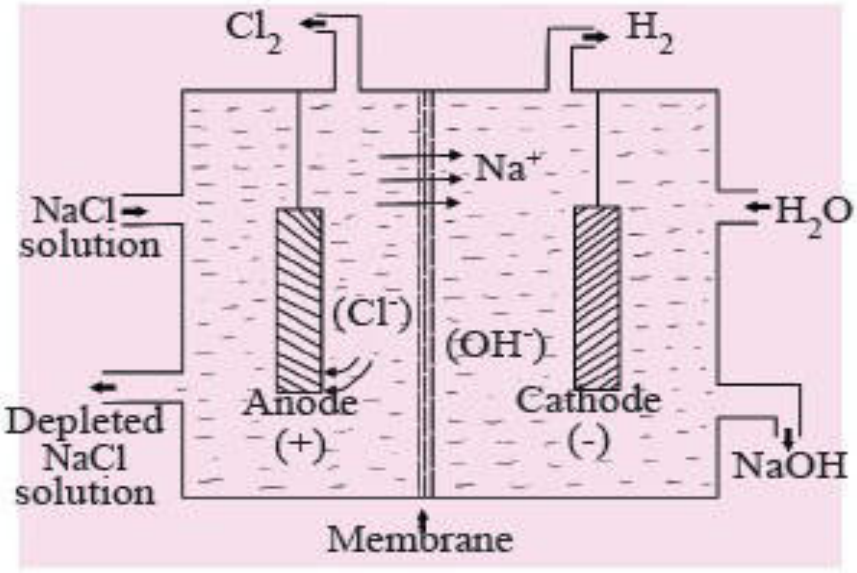
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Alkalies are substances which increase the concentration of hydroxide ions in aqueous solutions.

Arrhenius stated that when acids and bases dissolve in water, they dissociate into ions. According to Arrhenius theory, acids are substances which liberate H+ ions and bases are substances which liberate OH– ions in aqueous solutions.

Sodium hydroxide

Sodium hydroxide is industrially produced by chlor-alkali process. In this, concentrated aqueous solution of sodium chloride (brine) is subjected to electrolysis. In modern industries, membrane cell is used for the electrolysis of brine. A diagram of the membrane cell is given in Figure.



In this, the two chambers are separated using a membrane which allows the passage of specific ions only.

What are the ions formed when sodium chloride dissolves in water? .....

Which ion undergoes oxidation at the anode? .....

Complete the oxidation equation.

$2Cl^- \rightarrow$ ..... + ...

Which gas is liberated in the left chamber? .....

Na^+ ions in the left chamber move to the right chamber through the membrane.

Water molecules undergo reduction at the cathode in the right chamber.

Complete the reduction equation.

$$2H_2O + 2e- \rightarrow ..... + 2OH^-$$

Which gas is liberated here? .....

Sodium hydroxide is formed in the right chamber by the combination of Na^+ ions and OH^- ions.

$$Na^+ + OH^- \rightarrow NaOH$$

In this process, hydrogen and chlorine are also formed in addition to sodium hydroxide. Sodium hydroxide is also known as caustic soda.

Uses of sodium hydroxide

- For the preparation of soaps and detergents.
- For the purification of bauxite in the manufacture of aluminium.
- To produce paper.
- For the production of rayon.
- To manufacture medicines like aspirin.

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Hydrochloric acid (HCl)

What is the product obtained when hydrogen and chlorine react?

Write the balanced chemical equation. .....

Hydrogen chloride is manufactured by the direct combination of hydrogen and chlorine. Since it is an exothermic process, the reactor where the process takes place will always be hot. Hence the reactor is known as 'HCl oven' or 'HCl burner'. The hydrogen chloride formed in the HCl burner is dissolved in deionised water to produce concentrated hydrochloric acid.

Uses

- To refine table salt.
- To prepare dyes.
- For manufacturing PVC
- To prepare aqua regia which is used to dissolve metals like gold and platinum.
- For preparing organic and inorganic compounds.
- To produce fertilizers.
- For refining metals.

Salts

Which is the salt formed when sodium hydroxide reacts with hydrochloric acid?

.....

Examine whether the solutions of the salts given below are acidic or basic in nature and complete Table

Salt solution	Change in the colour of blue litmus paper when it is in contact with the solution	Change in the colour of red litmus paper when it is in contact with the solution	Nature of solutions (Acidic nature / basic nature / neutral)
NaCl	No change in colour	No change in colour	Neutral
NH_4Cl			
Na_2CO_3			

Certain salts when dissolved in water do not exhibit acidic or basic nature. Such salts are called neutral salts.

Salts that undergo hydrolysis when dissolved in water to give an acidic solution are called acid salts.

Eg. NH_4Cl, $(NH_4)_2\ SO_4$

Salts that undergo hydrolysis when dissolved in water to give a basic solution are called basic salts.

Eg. Na_2CO_3 , CH_3COONa

Salt hydrolysis is the process by which the anions, cations or both of a salt react with water to form a solution that has acidic or basic nature.

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- The salt formed by the reaction of a strong acid and a strong base will be neutral.
- The salt formed from strong acid and weak base will be acidic.
- The salt formed from weak acid and strong base will be basic.

Fertilizers

Natural or artificial substances that provide nutrients to plants are called fertilizers.
What are the elements present in water and carbon dioxide?

.....

These elements are available to plants from nature itself. They can be absorbed from air and water. These three elements are called natural nutrients.
Nitrogen (N), phosphorus (P) and potassium (K) are the three elements that plants need in large quantities. These are called primary nutrients.
The soil contains low amounts of calcium (Ca), magnesium (Mg) and sulphur (S). Plants need these only in small amounts. These three elements are called secondary nutrients. Some elements are needed to plant only in very small amounts. Hence, they are called micro nutrients.
Eg. Iron, manganese.

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Essential qualities of fertilizers

- The elements present in the compounds in fertilizer should be easily available to plants.
- The fertilizer should be soluble in water.
- The compounds in fertilizer must be stable and remain in the soil for a long time, making them available to plants.
- They should not cause significant variations to the pH of the soil.
- They should not be toxic to plants.

Chemical fertilizers are used to make the primary nutrients available to plants in the right quantities. Such fertilizers are classified as nitrogenous fertilizers, phosphate fertilizers and potash fertilizers. These are combined to produce mixed fertilizers. These are also called NPK mixed fertilizers.

Types of fertilizers	**Effect of fertilizer on plants**	**Examples**
Nitrogenous fertilizers	Accelerates the growth of plants.	Ammonium sulphate, calcium cyanamide, urea, sodium nitrate, calcium nitrate, potassium nitrate
Phosphate fertilizers	• Helps the growth of plants. • Increases productivity	Ammonium phosphate, triple super phosphate
Potash fertilizers	• Helps the growth of plants. • Increases productivity. • Boosts immunity	Potassium chloride (Muriate of potash), potassium sulphate, potassium – magnesium sulphate

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